

1. CT-500 Specification.

1. Core : ARM926EJ

- ▶ 8(4)KB Instruction Cache & 8(4)KB Data Cache.
- ▶ Operation Frequency : Core system clock(200MHz), Bus clock & SEIP clock(100MHz)
- ▶ S-AXI Bus
- ▶ Data Bus Internal 32bit / External 16bit.

2. Clock & PLL

- ▶ External : 27 MHz X-tal
- ▶ Core Clock, Video Clk(x), Sound Clk(?)

3. Memory & Controller

- ▶ Memory Controller – DDR, NAND, Static Memory, IO device(Low latency).
- ▶ DDR-SDRAM Controller(16bit).
 - 128/256/512 Mbit
- ▶ NAND Flash Controller(for 8bit).
 - 32M ~ 2G Byte.
 - 512/256 byte, 1-bit ECC.
 - Command FIFO/Data FIFO.
 - Support small/large block.
 - Support SLC/MLC.
- ▶ Extra Asynchronous Chip select signal for reservation : 4 (8/16bit).
 - Async IF – ex) ROM/Nor-flash/SRAM IF.
 - Wait configurable.
 - Upto 128M Byte.
- ▶ Internal SRAM(MCP option).
 - 4M bit.
- ▶ Mask ROM for Sound Engine(8bit).
 - 32/64/128Mbit.
- ▶ Internal SRAM for Sound Engine.
 - 1M bit + 24k bit.

4. Boot Sequence

- ▶ Boot seq. (selectable via IO pin)
- ▶ Internal Boot ram. (64k word – use SEIP effect ram)
- ▶ Direct Boot – from Wave ROM.
- ▶ NAND Boot.
- ▶ UART Boot.
- ▶ Boot Code makes decision, App. code runs on MCP SRAM or External DDR.

5. Sound Engine (Enter-tech)

- ▶ Operation Clock : 100MHz.
- ▶ (128K + 3072)Byte SRAM
 - Sound effect & MIC echo memory.
- ▶ 512Byte ROM.
- ▶ Audio CODEC : Excluded.
- ▶ Wave Playback & Recorder Interface.
- ▶ 1024 point FFT with I2S receiver.
- ▶ 4ch I2S Master interface : 2ch ADC in & 2ch DAC out .
- ▶ PCM Rx/Tx interface for Record & Play.

6. Graphic Controller

- ▶ NTSC/PAL output mode : 704 x 480i / 704 x 576i (max)
- ▶ VGA output mode : 640 x 480p(VGA), 320x240p(QVGA)
- ▶ 4pages
 - . Wide(1440x576) x 2page : no Palette
 - . VGA (640x576) x 2page : Palette 256 color/page
- ▶ Overlay & α -blending : 3pages(Wide 1 page + VGA 2pages).
- ▶ Support chroma keying : Color selectable.
- ▶ ~~Line Up-scaling : 480 $\rightarrow$ 576 (NTSC $\rightarrow$ PAL).~~
- ▶ Up-scaling support function
 - For helping NTSC- $\rightarrow$ PAL conversion.
 - Insert 1 horizontal line, copied from the 5th line, each 5 lines.
 - No line filter required.
- ▶ Configurable Outputs : RGB(565) & Sync/ITU-R BT.601(16bit)/ITU-R BT.656(8bit).
- ▶ Internal Color Bar Generator.
- ▶ Color Space Conversion.

7. Video Encoder

- ▶ NTSC/PAL display
- ▶ One DAC for **CVBS**, two times over-sampled with 10 bit resolution.
- ▶ Internal Color Bar Generator.
- ▶ Cross color reduction filter.

8. External Video Input

- ▶ Digital Video Decoder Interface : ITU-R BT.656(8bit).
- ▶ Supports Interlace / Non-interlace Mode.
- ▶ Color Space Conversion.
- ▶ Memory Buffering/~~Bypass~~ Mode.
- ▶ Support Frame Capture.

9. E-DMA

- ▶ 2 x 4ch + Graphic DMA x 1ch.
- ▶ Memory to Memory, Memory to IO.
- ▶ Block Copy (2-dimension)

10. Power Save

- ▶ Support sleep mode.
- ▶ Disabling module clocks.
- ▶ USB power Saving.
- ▶ DAC power control.
- ▶ PLL power down mode.

11. Peripherals

11.1 USB2.0 Device Controller

- USB 1.1/2.0 compliant.
- PHY included.

11.2 UART : 2ch + 2ch.

- Special UART (2ch)
 - Baudrate : 115200 max
 - Tx 32bytes & Rx 32bytes FIFOs.
 - 31.25k baudrate support for MIDI.
- Standard UART (2ch)
 - Baudrate : 115200 max
 - Tx 8bytes & Rx 8bytes FIFOs.

11.3 Timer : 4ch.**11.4 Watchdog Timer**

- 1m sec ~ 8 sec.

11.5 I²S : 2ch.

- Configurable Master/Slave.
- ADC(receive) & DAC(transmit) IF for audio Codec.
- IF mode : Standard I²S, MSB(=Left) justified mode(configurable).
- 8/16/24/32bit Word lengths.
- 16k/32k/44.1k/48kHz Sampling Frequency.
- Each 64 word FIFO entries for Transmit & Receive.

11.6 I²C Master

- Variable clock rate : 400kHz max.

11.7 SPI Master : 2ch

- SPI clock speed up to 12MHz.
- Ch-1 : For transferring MP3 bit-stream.
 - 160Byte/10mS @128k Sampling.
 - Support DMA.
- Ch-2 : For transferring commands.
 - Data Length(max) : 64bit(8byte).
 - No needs DMA.

11.8 Interrupt Controller

- Support 8 external interrupt sources.
- Priority programmable.

11.9 GPIO : 48 max.

12. Operating Voltage : 1.2V(core), 3.3V(IO), 2.5V(DDR IO)

13. Power Consumption : ? mW/MHz @1.2V (TBD)

14. Package : 256p or 272p fBGA or ?

15. Process : TSMC 0.13u CMOS

*, CT-500 Block Diagram

